

Name: _____



Securing understanding of sequences

Task A: Continuing sequences from a rule

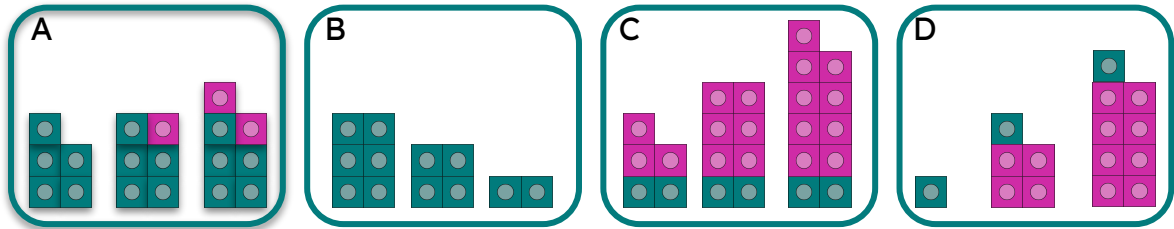
- 1) For each two starting numbers: continue the sequence with a constant additive pattern, then a constant multiplicative pattern. Describe each of your sequences in words.

	Additive pattern	Multiplicative pattern
a) Sequence	1, 3, __, __, __, ...	1, 3, __, __, __, ...
Description		
b) Sequence	0.1, 1, __, __, __, ...	0.1, 1, __, __, __, ...
Description		
c) Sequence	-16, 8, __, __, __, ...	-16, 8, __, __, __, ...
Description		
d) Sequence	0.5, 1.5, __, __, __, ...	0.5, 1.5, __, __, __, ...
Description		
e) Sequence	$\frac{1}{8}$, $\frac{1}{4}$, __, __, __, ...	$\frac{1}{8}$, $\frac{1}{4}$, __, __, __, ...
Description		
f) Sequence	25, 10, __, __, __, ...	25, 10, __, __, __, ...
Description		



Task B: Representations of sequences

Izzy has built these patterns using blocks. Which two patterns could she combine to make a sequence which



1) Increases by 5 each time

2) Increases by 2 each time

3) Decreases by 1 each time

4) Laura and Andeep are making a repeating pattern from different shaped counters



a) What shape counter will the number 25 be?

b) Laura puts down the counters 144, 145 and 146, what shapes are they?

c) Andeep puts down three counters in a row which are all oval shapes. They are all greater than 50 but less than 60, what numbers could they be?

d) Laura and Andeep run out of counters when they put down the counter 178. How many of each type did they use in total